



New vector and vaccine platforms: mRNA, DNA, viral vectors

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Purpose of review

The purpose of this review is to share the excitement of new developments in the field of vaccine vector modalities against infectious diseases. The focus is on HIV-1/AIDS with reference to the most successful as well as currently tested COVID-19 vaccines, and human trials, which best inform iterative vaccine improvements.

Recent findings

Several genetic subunit vaccines against SARS-CoV-2 demonstrated protection against severe disease, obtained Emergency Use Authorization and scaled their production to billions of doses. Many more are in efficacy evaluation. In contrast, development of HIV-1 vaccines has been extremely difficult. Perseverance of scientists is deepening our understanding of what constitutes immunity against HIV-1 infection and how to achieve protective levels of relevant responses by active immunization, passive administration or a combination of both. Novel platforms led by RNA play a pivotal role. However, a difficult virus may require a complex approach. Proof of concept for HIV-1 prevention and cure might be at reach, and when it arrives, it will be a great and needed encouragement to the field.

Summary

Despite the enormous success of drug treatment, vaccines remain the best solution and likely a necessary component of any package that truly ends the AIDS epidemic.

Keywords

ChAdOx1, clinical trials, DNA vaccines, HIV cure, HIV prevention, RNA vaccines, subunit genetic vaccines

INTRODUCTION

In the U.S., there are currently 28 pathogens with a licensed vaccine [1]. Many more are still awaiting an effective vaccine strategy. For more complex diseases such as AIDS, tuberculosis (TB), malaria, cancers, autoimmune and neurodegenerative diseases, vaccine development requires a much greater understanding of the principles and pathways towards effective immunity. Intensive research into HIV-1 vaccines is entering its fifth decade, although with fewer than 10 funded phase 3 trials testing preventive efficacy of candidate vaccines since the start of the AIDS epidemic (compared with nine licensed vaccines and over three dozen vaccine efficacy trials in the first 2 years of the COVID-19 pandemic [2]). Regrettably, the few trialled HIV-1 vaccine strategies have failed to show deployable, reproducible efficacy [3–6]. The lack of efficacy of the candidate vaccines tested to date have been due to product failures [7] rather than to the intrinsic impossibility to elicit actively protective levels of broadly neutralizing antibodies [6] or that robust, broad and well matched killer T cells could not

control HIV-1 [8,9]. The devil is in the vaccine details. Conceptually, vaccines have several key components, which together ensure generation of protective responses, and any one of these components performing suboptimally can render the whole vaccine strategy ineffective [10]. These include at least the HIV-1-derived immunogens, their delivery, vaccine formulation and immunomodulation [11,12,13]. Although some argue that HIV-1 research is now an immunogen discovery

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KEY POINTS

- Even in the broader context of increasing antiretroviral treatment and prevention, HIV-1 vaccine remains the best solution and likely a key component to any package truly ending the AIDS epidemic.
- Vaccines against difficult diseases such as AIDS will require deeper understanding of the pathways towards protection and demand more complex solutions likely harnessing the protective potentials of both antibody and T-cell responses.
- Novel vaccine vectors/platforms led by RNA will enable and accelerate vaccine development; following the COVID-19 success, scientists are eager to validate the RNA platform for other diseases.
- The importance of vaccine development and preparedness cannot be overestimated.
- Scientific success of vaccine development must be backed by scalable manufacture to meet the global demand, reaching people who need the vaccines the most and increasing the trust of the communities as well as local engagement and leadership to maximize the vaccine uptake.

problem, the importance of other components is not to be underestimated.

Vaccines induce host immune responses through a complex set of interactions with multiple cell types, innate immune receptors and pathways, and the mechanisms required for protection vary from pathogen to pathogen. Furthermore, individuals develop different ways of controlling the same virus depending on their genetic makeup [14,15], abundance and molecular state of their preexisting immune cells, including history of other infections offering cross-reactivity and their unique encounter with HIV-1 [16]. The Holy Grail of HIV-1 vaccinology is the induction of broadly neutralizing antibodies. Given the challenges bnAb induction faces and the clear evidence that T cells impose a selective pressure on HIV-1 during natural infection [17], effective vaccines should harness killer T cells if only to complement bnAbs [8,17]. Killer T cells might be particularly important for cure because they are better at killing infected cells than antibodies. In turn, antibodies may enhance T-cell responses through a ‘vaccinal effect’ [18]. Clearly, vaccines must induce better immune responses than natural HIV-1 infection.

In this article, I provide a brief overview of the challenges and opportunities provided by the RNA, DNA and viral vector platforms delivering HIV-1 immunogens.

RAPIDLY EVOLVING LANDSCAPE OF THE RNA VACCINE PLATFORM

Vaccines based on RNA are the ‘hottest’ vaccinology development with the momentum and potential to transform the field [11[■],19–21]. However, RNA alone will not solve all the current vaccine roadblocks for the most difficult diseases, because which immunogens RNA expresses and how it is formulated also matters [22]. Nevertheless, RNA can serve as a tool for rapid iterative optimization of vaccines in humans and significantly accelerate the rate of progress. Intramuscular RNA injection as a strategy for protection against or cure for a variety of cancers and infections including HIV-1 has been in development since late 1980s. Initial obstacles of instability, inefficient delivery, activation of innate immunity through intracellular RNA sensors and large-scale manufacturing have been overcome sufficiently to allow Moderna’s mRNA-1273 and BioNTech/Pfizer’s BNT162b2 vaccines to emerge during the COVID-19 pandemic as well tolerated and efficacious vaccines [23,24]. At the time of writing this article and as a reflection of the interest in this platform, 37 other RNA-based COVID-19 vaccines are in clinical development representing 22% of the total global COVID-19 vaccine effort [2].

The RNA platform is a type of genetic vaccines. Instead of delivering immunogenic proteins, genetic vaccines provide the protein’s coding sequence, which instructs the host cells how to produce the vaccine immunogens. RNA vaccines can be delivered directly as messenger RNA (mRNA) [19] and as a self-replicating RNA (srRNA) [11[■]]. mRNA vaccines have smaller molecules and can be used in homologous boosts without reduced efficacy due to antivector immunity. The advantage of srRNA is that after entering cells, it replicates and increases the copy number of the mRNA. This translates into higher immunogen production leading to improved immunogenicity and ultimately lower vaccine doses. srRNAs are purpose engineered replicons not only derived mostly from alphaviruses Venezuelan Equine Encephalitis, Sindbis and Semliki Forest viruses, but also from flaviviruses, nodamura viruses and nidoviruses. Although the successful COVID-19 vaccines of Moderna and BioNTech were both nonreplicating mRNAs, srRNA features are attractive and its performance in humans is improving [11[■]].

Design of the mRNA nucleotide sequence begins with the decision on the immunogen. This is typically coupled to a strong N-terminal leader sequence, and for HIV-1 Env contains various intramolecular as well as trimer-stabilizing and glycosylation-finetuning amino acid modifications. The ‘know-how’ in the designing of the mRNA molecule

itself is nontrivial. The RNA molecule starts with a modified dinucleotide GA (guanine-adenosine), called the 5' cap, which enhances mRNA stability by impacting on innate sensing. Both the 5' and 3'-untranslated regions (UTRs) contain critical regulatory elements [21,25,26] and can be derived from highly expressed human genes or use the pathogen's original mRNA UTRs assuming optimization by natural evolution. Translation initiation is often the rate-limiting step in protein synthesis and begins from the first methionine start codon enhanced by a modified Kozak consensus sequence and its flanking secondary RNA structure directing the ribosome subunit to the first AUG. One advantage of natural UTRs is that they are almost certainly devoid of hidden upstream start codons. Taking the publicly available BioNTech/Pfizer SARS-CoV-2 vaccine BNT162b2 sequence [27] as an illustration, the 5'-UTR contains 52 nucleotides, which originate from the highly expressed human α -globin gene. Codons are optimized according to highly expressed human genes, which not only makes them generally GC-rich, but also for the accuracy of translation [28¹¹]. This, together with a higher availability of corresponding amino acid charged transfer RNAs, leads to more efficient protein expression. An interesting safety feature is the reduction of the protein-coding potential by avoiding short overlapping ORFs on both the positive and, for srRNA, negative RNA strands. Notably, mRNA-1273 eliminated all overlapping ORFs on the positive strand compared with eight shorter ORFs found on BNT162b2 and 11 on the wild-type SARS-CoV-2 spike gene [29]. The main ORF ends with two (BNT162b2) or three (mRNA-1273) consecutive stop codons to compensate for pseudouridine (see below) mispairing. The 3'-UTR follows, which is critical for mRNA localization, stability and translational efficacy. The 136 nucleotides of the BNT162b2's 3'-UTR is like the amino-terminal enhancer of split (AES) mRNA improved by computation to minimize miRNA binding. The 3' end of mRNA is polyadenylated, which further protects against degradation. The optimal length of polyA is approximately 120 A's. The BNT162b2 vaccine ends with 30 A's, a 10-mixed nucleotide spacer and a further 70 A's, whereby the mixed-nucleotide spacer facilitates the manufacturing process involving the template plasmid. Several software assist the mRNA designs by analysing the ORF for codon frequencies, codon adaptation index, index of translation efficiency, position weight matrix and minimum folding energy [28¹¹].

RNA is inherently immunostimulatory to the innate immune system through interaction with pattern-recognition receptors. This leads to the production of type I interferon and pro-inflammatory

cytokines. Type I interferon response matures dendritic cells but can be also counterproductive by shutting down the protein synthesis (including that of the immunogen), altering or interfering with protein processing and presentation, affecting autophagy and inducing cellular stress response. All these processes influence the vaccine immunogenicity and reactogenicity. To decrease innate stimulation, RNAs often employ modified nucleosides. Both BNT162b2 and mRNA-1273 replaced uracil with *N*₁-methyl-pseudouridine, but other modifications such 5-methylcytidine can be also used [30,31]. However, the pairing of modified nucleotides is less accurate. Importantly, responses to modified and unmodified mRNA differ between mice, nonhuman primates and humans, thus fine tuning of the vaccine is best done in the relevant species.

Once the mRNA molecule is designed and several mRNAs can compose a single vaccine, it is produced and purified comprehensively from dsRNA and other manufacturing impurities [32]. For efficient delivery, RNA must be protected from degradation by a lipid formulation, which can have a form of liposomes, lipoplex, lipid nanoparticles (LNPs) or polymeric nanoparticles [11¹,19–21]. A typical LNP contains four important excipients: cationic lipids neutralizing RNA's negative charge or a new generation of ionizable amino lipids, which change charge in acidified endosomes to facilitate the RNA dissociation and endosomal exit into cytoplasm, a helper phospholipid supporting the lipid bilayer, cholesterol adjusting the membrane fluidity and polyethylene glycol-lipid improving colloidal stability and shielding from opsonization. The formulation must be of minimum toxicity, which is greatly increased by biodegradable linkages [33]. It also facilitates several important features such as membrane crossing, which may employ cell-penetrating peptides, or specific cell-targeting by attachment of ligands. Cytoplasmic delivery of naked RNA can be increased by electroporation. Every new product will likely need optimization by preparing a few starting RNA variants and formulations to select empirically the most immunogenic blend. BioNtech tested five COVID-19 vaccine candidates to select BNT162b2.

Candidate RNA constructs for HIV-1 were used initially to pulse monocyte-derived dendritic cells, which were transferred back into patients. This strategy showed moderate immunogenicity but offered no or very limited viraemic control [34]. The first HIV-1 mRNA vaccine study was launched in 2015 and was a phase 1 dose-escalation trial in chronically infected patients stably controlling HIV-1 by ART. Naked HIVACAT T-cell immunogen (HTI)

[35] mRNA adjuvanted with TriMix mRNAs encoding two dendritic cell activating proteins cluster of differentiation (CD)40L and caTLR4, a constitutively active form of TLR4, as well as T cell-activating CD70 [36] was delivered intranodally and was well tolerated (NCT02413645) [37]. The follow up phase 2a trial showed no immunogenicity, no decrease in latent HIV-1 reservoir and no post-ART control in part due to mRNA design error (NCT02888756) [38]. In November 2021, LNP mRNA vaccines eOD-GT8 60mer (mRNA-1644) and Core-g28v2 60mer mRNA (mRNA-1644v2-Core) self-assembling into virus-like particles entered clinical evaluation and are tested individually and in combination in HIV-1-negative volunteers (NCT05414786). The aim of the eOD-GT8 immunogen is to engage the unmutated germ-line B-cell precursor with the potential to evolve through somatic mutations into a broadly neutralizing glycan-dependent VRC01-class antibody [39]. IAVI G002 followed in May 2022 delivering eOD-GT8 HIV-1 60mer mRNA to HIV-1 negative African adults (NCT05001373). Finally, in March this year, a phase 1 trial of HIV Vaccine Trials Network (HVTN) 302, opened recruitment administering three variants of HIV-1 Env trimers BG505 MD39.3, BG505 MD39.3 gp151 and BG505 MD39.3 gp151 CD4KO expressed from encapsulated mRNA and given to healthy HIV-1-negative volunteers at multiple sites in the USA (NCT05217641). All the Env LNP mRNA vaccines were produced by Moderna and the results from these studies are expected in 2023. For HIV-1, the landscape of mRNA vaccines will evolve rapidly.

MAJOR ADVANCEMENTS FOR DNA VACCINES ARE STILL AWAITED

DNA began development as a vaccine modality at approximately the same time as RNA [40] and spawned tremendous expectations [41]. The big difference for DNA is that it must cross the cytoplasmic as well as nuclear membranes for expression. The transport of transcribed RNA to cytoplasm for translation can be enhanced by the addition of splice donor and acceptor sites, but caution must be taken not to inadvertently introduce other splicing signals within the 5′-/3′-UTR and ORF design. DNA has been immunogenic in mice and a few other preclinical models, but its immunogenicity in humans remains moderate at best. Up to 8-mg doses, codelivery with cytokines and chemokines [42], biojector devices [43], DNA-coated gold particle intradermal delivery (gene gun) [44], intradermal or intramuscular electroporation [42,45], tattooing [46], magnetoporation/sonoporation/optoporation [47], transcutaneous microneedle strategies [48], dermavir [49], depletion of normal human plasma

major DNA-binding protein [50] are either awaiting human testing or have so far failed to deliver a major breakthrough of the DNA platform for human use [7,51]. Moderate improvements by co-delivery of interleukin (IL)-12 and electroporation in trial HVTN 098 (NCT02431767) were observed [52], but these are still easily dwarfed by other delivery modalities. As a prime, DNA increased responses following viral vector boost [45,53–55]. It is interesting whether in this setting, the low immunogen expression following plasmid DNA priming improves the quality and durability of responses after the viral vector boost [45,56]. Currently, there are 16 DNA vaccines in clinical testing against COVID-19 representing 10% of the global COVID-19 vaccine effort [2].

THE POTENTIAL OF VIRAL VECTORS IS IN THEIR DIVERSITY

Development of well tolerated purpose-designed viral vectors is extremely important. There is an unmet need for engineered vaccine vectors and modalities to cover the global demand for safer and more efficient subunit vaccines than those currently in use, against difficult pathogens such as HIV-1, and all the identified and yet unknown pandemic threats. The aim is to develop a panel of well tolerated viruses, which can be administered not only as stand-alone vaccines, but also in heterologous prime-boost regimens to complement and enhance each other, provide sufficient options to rescue first-line vaccine nonresponders and ensure efficient long-term maintenance of immunity by alternating heterologous boosts [8]. Vaccine vectors most commonly under development include those derived from human adenoviruses of serotypes 4, 5, 26 and 35 [4,57,58], simian adenoviruses isolated from chimpanzees [8,59] and gorillas [60,61], vesicular stomatitis virus (VSV) [62], adeno-associated viruses (AAV) [63], integration-deficient lentiviruses (IDLV) [64,65], poxviruses such as vaccinia virus, modified vaccinia Ankara (MVA), fowl poxvirus and human cytomegalovirus (HCMV) to name a few. Vector platforms have been comprehensively reviewed many times over, most recently in a well illustrated review by Travieso *et al.* [66]. The first viral vector that was given Emergency Use Authorization and the most widely globally used vaccine against SARS-CoV-2 is vectored by ChAdOx1, which was engineered from a simian adenovirus Y25 isolated from a chimpanzee [67]. It is replication deficient, whereby for the manufacture, the essential missing function must be supplied *in trans* by a stably transformed cell line [68]. These cells also produce a tetracycline repressor slowing

substantially the transgene transcription and therefore lowering the biological burden on the vaccine-producing cells during manufacture; this in turn increases vaccine yield and stability. For transgenes, care must be taken not to introduce unwanted splicing events, which can produce reactogenic products. Splicing prediction is inherently unreliable; thus, experimental testing may become a standard [69[■],70]. ChAdOx1 is explored by two candidate HIV-1 vaccines delivering conserved mosaic (HIVconsvX) [59] (NCT04586673 and NCT04553016) and protective (HTI) [35] (NCT03204617) T-cell regions. Nonreplicating and replicating virus vectors represent 21 (13%) and 4 (2%) of the current anti-SARS-CoV-2 vaccine strategies in clinical testing, respectively [2].

Virus vectors come in many different 'shapes and forms'. Vectors can carry single or double stranded nucleic acids, RNA or DNA, a single positive or negative RNA strand, be enveloped or non-enveloped and simple or complex (expressing from a few to over 200 viral proteins). Some vectors can replicate in the host, and others are replication deficient. Replication makes vaccines more immunogenic but requires extra safety vigilance. Vector tropism is dictated by the ligands for cell entry receptors viruses employ. Specific surface markers can be explored for targeting vaccines to professional antigen-presenting cells for efficient induction of immune responses [65]. Viral RNA, DNA and also other pathogen-associated molecular patterns intrinsic to vectors are a powerful stimulus to the innate immune system. Like RNA vaccines, innate pathways can enhance immunogenicity, but also impact on protein expression and presentation and bias the stimulation of adaptive immune responses. Hence, not all viral vector-based vaccines will necessarily lead to protective responses for all immunogens and against all pathogens. This challenges the plug-and-play or one-size-fits-all aims of vaccine platforms for pandemic preparedness and underlines the need for a panel of alternative platforms. Two examples are AAV and VSV. AAV prior to vaccine engineering was poorly immunogenic due to inefficient uncoating in dendritic cells and so was more suitable for gene therapy [71]. VSV was only weakly immunogenic for the SARS-CoV-2 insert but is licensed as a vaccine against Ebola [62]. For HIV-1 prevention, vaccines stimulating de-novo responses may require a strong pro-inflammatory stimulation, but once the control of HIV-1 is established, therapeutic vaccines aiming to achieve post-ART HIV-1 remission may benefit from more 'silent' non-inflammatory vectors [72]. At the same time, strategies reactivating latent provirus reservoir must activate NF- κ B, because

HIV-1 transcription is strictly dependant on this transcription factor.

The best example of unique fine-tuned protection against a pathogenic SIV challenge is the consistent protection of approximately 55% of rhesus macaques achieved by particular molecular clone 68-1 of rhesus cytomegalovirus (rCMV) expressing SIV immunogens [73]. This vector carries five specific deletions in the viral genome, which completely prevent classical macaque major histocompatibility complex class Ia Mamu-A, B and C presentation of epitopes and instead induce non-classical class Ib Mamu-E and class II-restricted CD8⁺T cells. The mechanisms of this protection are being gradually deciphered [74[■],75,76], but it is species specific, as the same rhesus CMV vaccine does not protect cynomolgus macaques. Engineering of its human CMV equivalent has been attempted and candidate vaccine VIR-1111 is currently being evaluated in a phase 1 clinical trial in humans (NCT04725877).

CONCLUSION

Development of an effective vaccine against the most difficult of viruses, HIV-1, has been extremely challenging. The COVID-19 pandemic has momentarily focused the minds of politicians and funders on vaccinology and HIV-1 can benefit from this just like the SARS-CoV-2 vaccines benefited from decades of HIV-1 research. Novel vaccine vectors for subunit vaccines are an essential component of eventual success, which is now more in our grasp than it has ever been before.

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Conflicts of interest

The author is one of the inventors on EU patent EP14846993.5 and U.S. patent PCT/US14/58422 concerning the HIVconsVX immunogen.

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- of special interest
- of outstanding interest

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